

Supplementary Material: State-Gated Promotion for Multilingual LoRA Adapters

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1 Released Audit Assets

The submission source includes the versioned probe files, automatic scorers, and raw audit tables needed to regenerate the G2-52, G3-80, and four-language main-boost summaries. The released source tree contains:

- fae_protocol/probes_v2_translit.jsonl: 52 G2 script-state prompts.
- fae_protocol/probes_v3_schema.jsonl: 80 G3 structured-output prompts.
- fae_protocol/score_translit_auto.py: Unicode-block script-state scorer.
- fae_protocol/score_schema_auto.py: JSON/schema scorer.
- figures/g2_extended_scores.json and figures/g3_extended_scores.json: merged audit results.
- fae_protocol/probes_v4_4l_audit.jsonl: 112 four-language KO/RU/FR/EN audit prompts.
- figures/audit4l_main_boost_scores.json: merged four-language gate audit results.
- figures/common_4l_main_boost_loss.json: common held-out loss for the four-language seeds.
- figures/app_constraints_4l_summary.json: app-constrained deployment action summary.
- audit4l_main_boost_raw_generations.jsonl: raw outputs for the main-boost audit (in figures/).
- figures/audit4l_repair_scores.json: single-arm curriculum decomposition results (curriculum decomposition table in the main paper).
- figures/audit4l_parity_fp16.json: FP16-inference parity check on the seed-10 winner.
- Stock-baseline gate audits for the cross-base comparison: figures/audit4l_stock_gemma4_e4b.json, ..._qwen25_3b.json, ..._llama32_3b.json, ..._phi35_mini.json (an additional bloomz_3b audit is included for completeness but is not used in the main-paper table).
- figures/audit4l_summary.json: per-variant gate summary including the Gemma 4 E2B stock reference row used in the main-paper cross-base table; figures/audit4l_scores.json contains the corresponding per-probe scores.

- Policy+family cross-base runs: figures/audit4l_cross_qwen.json, ..._qwen_s1500.json, ..._llama.json, ..._llama_s1500.json, ..._phi.json, ..._phi_s1500.json, ..._gemma_ctrl.json, ..._gemma_ctrl_s1500.json. Unsuffixed = early audited checkpoint; _s1500 = final audited checkpoint (actual step counts differ by base; see main paper table caption).
- G2/G3 threshold-sensitivity tables corresponding to the engineering triage bands in the main paper: figures/g2_threshold_sensitivity.md, figures/g2_extended_selector_table.md, figures/g3_extended_selector_table.md.

2 Gate Definitions

G1 checks family-card structural and age-policy compliance. It is not a pedagogical-quality score. A response may pass G1 when it obeys the requested mode, age band, active languages, materialized language fields, and age-policy constraints.

G2 checks script-state compliance. It uses Unicode-block ratios and lightweight closed-set content sanity checks in the four-language audit. It is not a transliteration-quality benchmark.

G3 checks the app-parser schema contract. The scorer extracts one JSON object and then checks required keys, field types, enum values, forbidden extra keys, active languages, and age-policy fields. It is therefore an *extractable-schema* gate rather than a strict raw-JSON-only gate. A product path that requires raw JSON only should replace this scorer with a raw-only parser or constrained decoding.

G4 checks session routing. For a KO/RU/EN session, French leakage is disallowed; for a KO/FR/EN session, Russian leakage is disallowed. The scorer checks the declared active_languages, card fields, and full-output inactive-language markers.

3 Four-Language Main-Boost Results

The four-language sweep compares a combined policy+family repair curriculum against a no-policy ablation. Across three policy+family seeds, common loss is 0.6673 ± 0.0001 , G3 schema is $95.8 \pm 7.2\%$, and G4 session routing is $91.7 \pm 14.4\%$. Across two no-policy seeds, common loss is 0.8040 ± 0.0038 and both G3 and G4 are $0.0 \pm 0.0\%$. The app-constrained deployment summary yields two GREEN and one AMBER policy+family seeds, versus one GREEN and one RED no-policy seeds.

4 Single-Arm Curriculum Ablations

The main comparison uses repeated seeds for the combined policy+family repair curriculum and the no-policy ablation. Earlier single-arm runs provide additional diagnostic context. They are not treated as seed-averaged estimates, but they show why the combined repair is needed: improving schema alone does not guarantee session routing, and low loss alone does not make an adapter promotable.

The ablation pattern is useful because it rules out a simple “add any extra data” interpretation. Some low-loss variants still fail G4 entirely, and family-only or policy-only repair does not jointly recover family structure, schema, and session routing. The combined policy+family curriculum is not claimed to be optimal, but among the curricula reported here it is the only one whose effect on both common loss and the deployment-state gates is verified across multiple seeds (the single-arm variants are one-run diagnostics).

5 Closed-Loop Interpretation

The main contribution is the closed-loop trace: frozen gates define failure diagnosis, the failed states define a targeted policy+family repair curriculum, and the repaired adapters are re-evaluated under the same frozen gates. The individual ingredients—JSON validation, script checks, and targeted data—are familiar. The evidence is that connecting them in this loop changes both scalar loss and deployment-state behavior relative to the no-policy ablation.

6 Claim-Tier Policy

The paper separates three claim tiers. Tier 1 existence claims identify a concrete failure in a named artifact. Tier 2 predictability claims require controlled reproduction under a specified

training factor. Tier 3 prevalence claims require an independent population sample. The present submission makes a bounded four-language repair claim: Tier 2 under matched training settings on Gemma 4 E2B (matched on base/hyperparameters/audit/translation corpus, not on data volume) (3 policy+family seeds and 2 no-policy seeds); Tier 1 single-run stock-failure replication on four other bases (Gemma 4 E4B, Qwen 2.5 3B, Llama 3.2 3B, Phi-3.5 mini); Tier 1 single-run *repair* replication on three of those (Qwen, Llama, Phi). It does not make a Tier 3 prevalence claim.

7 Build Note

The main paper PDF is compiled from `source/lora_state_gated.main.tex` with the ACL style.

Group	n	Loss	G1	G1 struct.	G1 age	G3	G4
No-policy	2	0.8040 ± 0.0038	12.5 ± 17.7	35.4 ± 50.1	68.8 ± 26.5	0.0 ± 0.0	0.0 ± 0.0
Policy+family	3	0.6673 ± 0.0001	55.6 ± 6.4	62.5 ± 18.2	93.1 ± 12.0	95.8 ± 7.2	91.7 ± 14.4

Table 1: Four-language main-boost aggregate. Values are mean $\pm$ sample standard deviation across seeds; gate values are percentages.

Variant	G1	G2	G3	G4/action
No-policy (single seed)	41.7	85.0	20.8	0.0 / RED
Policy-only	4.2	95.0	100.0	0.0 / RED
Family-only	8.3	92.5	95.8	83.3 / RED
Balanced repair	25.0	95.0	100.0	33.3 / RED
Policy+family repair (mean)	55.6	91.7	95.8	91.7 / Free: 1 AMBER, 2 RED; Deployed: 2 GREEN, 1 AMBER
No-policy (mean)	12.5	92.5	0.0	0.0 / Free: 2 RED; Deployed: 1 GREEN, 1 RED

Table 2: Diagnostic single-arm curriculum ablations. Gate values are percentages. The single-arm rows are one-run diagnostics; only the policy+family and no-policy rows are repeated-seed summaries.

Variant	Gate	Output sketch	Scorer diagnosis
No-policy seed10	G3	Extractable JSON, but <code>age_band</code> is “0–2 months”, <code>card</code> is a string, and <code>safety</code> is a string.	Schema fails: wrong enum and wrong field types, even though a JSON object can be extracted.
No-policy seed10	G4	Translation-style object with <code>word</code> and <code>translations</code> , but no routing fields.	Required routing fields are missing: <code>mode</code> , <code>age_band</code> , <code>active_languages</code> , <code>card</code> , <code>next_action</code> , and <code>safety</code> .
Policy+family seed10	G3	Family-card object with correct <code>mode</code> , <code>age band</code> , <code>active languages</code> , <code>card</code> object, <code>next action</code> , and <code>safety</code> object.	Passes required keys, enum values, field types, and active-language constraints.
Policy+family seed10	G4	Parent-bridge object declares KO/RU/EN as active and instructs the interface to use only those languages.	Passes session routing: exactly the active KO/RU/EN languages are declared and inactive French is absent.

Table 3: Representative raw-output audit examples. The full raw generations are released in `audit4l_main_boost_raw_generations.jsonl`.